SIMATS Online Programming Lab

PYTHON PROGRAMMING

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A mathematical analysis system needs to verify whether a given number is prime. A prime number is a number greater than 1 that has no divisors other than 1 and itself.

Write a Python program that:

- Accepts an integer input from the user
- Checks whether the number is prime using appropriate control flow constructs
- Displays a suitable message indicating whether the number is prime or not

Ensure proper use of loops and conditional statements.

Your Code

```python
n = int(input())
if n<=1:
    print("not a prime number")
else:
    prime = True
    for i in range(2,n):
        if n % i == 0:
            prime = False
            break
    if prime:
        print("prime number")
    else:
        print("not a prime number")
```

Input

```
5
```

Output

```
prime number
```

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A numerical computation module is designed to generate a sequence where each number is the sum of the two preceding numbers. This sequence is known as the Fibonacci series.

Write a Python program that:

• Accepts a number n from the user

• Generates and displays the first n terms of the Fibonacci sequence

• Uses looping constructs to compute the sequence

Ensure the program handles cases where n is less than or equal to zero.

Your Code

```
n = int(input("enter the nunber "))
a=0
b=1
for i in range(n):
    print(a,end=" ")
    c=a+b
    a=b
    b=c
```

Input

```
5
```

Output

```
enter the nunber 0 1 1 2 3
```

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Q3 **String**

10 marks

A text processing system receives user input in the form of a string that may contain alphabets, digits, and special characters.

Write a Python program that:

• Accepts a string input from the user

• Counts the number of vowels, consonants, digits, and special characters present in the string

• Displays the count for each category

Use appropriate string operations and control flow statements.

Your Code

```
a = input("enter the number")
countvowels=0
countconsonants=0
countdigits=0
countspecial=0
for i in (a):
    if( i in "aeiouAEIOU "):
```

Input

```
74328yw89r
```

Output

Your Code

```
a = input("enter the number")
countvowels=0
countconsonants=0
countdigits=0
countspecial=0
for i in (a):
    if( i in "aeiouAEIOU "):
        countvowels = countvowels+1
    elif i.isalpha():
        countconsonants = countconsonants+1
    elif i.isdigit():
        countdigits=countdigits+1
    else:
        countspecial= countspecial+1
print("vowles",countvowels)
print("consonants",countconsonants)
print("digits",countdigits)
print("specialcharacter",countspecial)
```

Input

```
74328yw89r
```

Output

```
enter the numbervowles 0
consonants 3
digits 7
specialcharacter 0
```

Submitted

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A program is designed to check whether a number entered by the user is even or odd. However, the program contains syntax and logical errors.

Analyze the given code and perform the following:

- Identify all errors present in the program
- Rewrite the corrected program
- Briefly justify each correction

num = input("Enter number: ")

if num % 2 == 0

print("Even")

else

print("Odd")

Your Code

```
a = int(input("enter the number\n"))
if (a % 2 ==0):
    print("even number")
else:
    print("odd number")
```

Input

5

Output

```
enter the number
odd number
```

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Q5 **Print numbers from 1 to 10**

10 m

A program is intended to display numbers from 1 to 10 using a loop. However, some parts of the code are missing.

Complete the program by filling in the blanks so that it correctly prints numbers from 1 to 10.

Program to print numbers from 1 to 10

for i in ________:

print(________)

Also explain how the range() function works in this context.

Your Code

```
# program to print numbers 1 to 10
for i in range (1,11,1):
    print(i,end = " ")
#how the range function works means the range(a,b,c)
#which a is starting value including it also
#b is the ending value excluding it means if we give b = 10 then the p
#c is step value if step value is 1 it will increases gradually 1
```

Input

stdin input

Output

```
1 2 3 4 5 6 7 8 9 10
```